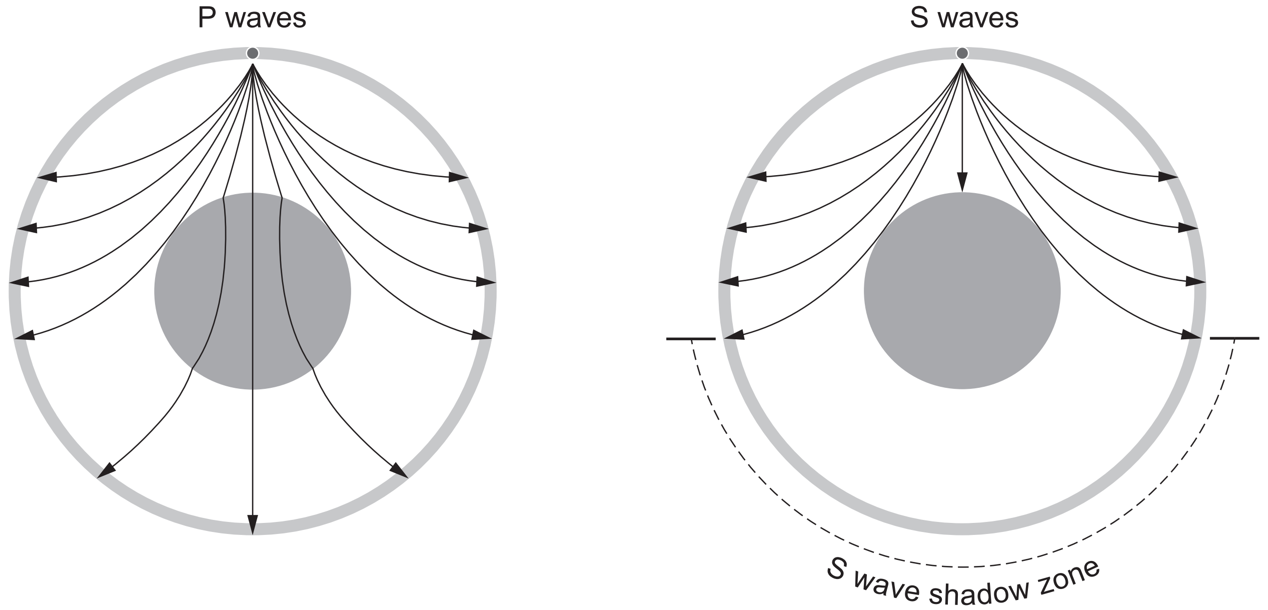
Examiner only

2. The diagram shows the path of seismic P and S waves through the Earth.



(a) (i) Underline the correct word in the bracket. [1]

The Earth has a liquid (**core** / **mantle** / **crust**).

(ii) Tick (✓) the **three** correct statements about P and S waves. [3]

P waves are transverse waves.	☐
P waves travel faster than S waves.	☐
S waves cannot travel in a liquid.	☐
P waves cannot travel in a solid.	☐
S waves are transverse waves.	☐
P waves cannot travel in a liquid.	☐

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(b) (i) P waves from an earthquake, which are travelling at a mean speed of 8000 m/s, reach a seismic monitoring station 500 s later.

Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the P waves to the monitoring station. [2]

distance = ... m

(ii) **A**, **B** and **C** are monitoring stations in Wales.
They collect data from an earthquake.

The epicentre is located somewhere on the circles shown in the diagram below.
Mark with a cross (x) the location of the epicentre. [1]

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07

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